

How satellite images work? A sketch of reflectance

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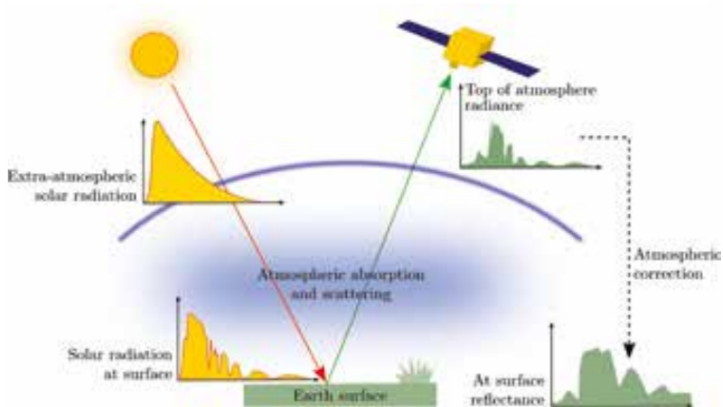
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Outline

- 1 How satellite images work
- 2 Radiant flux and reflectance
- 3 Bits

How satellite images work



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Radiant flux and reflectance

- Radiant flux (flow of energy): Φ (Watt)
- Reflectance: ratio between the reflected radiant flux (energy) and the incident radiant flux (energy):

$$\rho_{\lambda} = \frac{\Phi_{reflected}}{\Phi_{incident}} \quad (1)$$

Bits

The primary purpose of storing data in DN is to reduce file size.

Let's see how bits work...

Shannon information: 1 bit = 0 or 1.

Let's see how bits work...

- 2 bits, how many possible combinations?
- 00, 01, 10, 11
- 4 possible values

Let's see how bits work...

- 3 bits, how many possible combinations?
- 000, 001, 010, 100, 110, 101, 011, 111
- 8 possible values

What's the rule?

- 2^{nbit}
- $2^1 = 2$ values [0,1]
- $2^2 = 4$ values [00, 01, 10, 11]
- $2^3 = 8$ values [000, 001, 010, 100, 110, 101, 011, 111]



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